

Sean Luka Girgis

Senior Performance & Capacity Engineer | Forecasting, Telemetry Pipelines & Executive Reporting

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Professional Summary

Senior Performance and Capacity Engineer with 20+ years of enterprise technology experience and deep Citi-scale background in infrastructure telemetry, predictive capacity forecasting, KPI reporting, performance analytics, operational playbooks, and executive-facing insights. Built Python/Pandas/PySpark data pipelines ingesting telemetry from 6,000+ endpoints and transforming it into validated reporting, forecasting, and decision-support datasets. Strong fit for capacity engineering roles requiring large-scale infrastructure analysis, resource optimization, risk mitigation, budgeting support, operational readiness, stakeholder communication, and data-driven forecasting models.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Owned capacity and performance analytics workflows for large-scale banking infrastructure, transforming telemetry from 6,000+ endpoints into executive-ready insights for planning, risk reduction, and resource optimization.
- Built Python/Pandas/PySpark pipelines ingesting BMC TrueSight/TSCO, CA Wily, AppDynamics, Oracle, and AWS-backed telemetry into curated reporting and forecasting datasets.
- Developed ML forecasting workflows using Prophet and scikit-learn to project capacity trends, identify bottleneck risk, and support proactive infrastructure planning.
- Designed KPI-based reporting models and dashboards that converted raw utilization, performance, and capacity telemetry into actionable decision-support views for senior stakeholders.
- Supported capital planning and resource-allocation decisions by analyzing utilization patterns, underused infrastructure, seasonal demand, growth trends, and provisioning risk.
- Created operational runbooks, metric definitions, data flow notes, validation checkpoints, and support procedures to improve repeatability, team onboarding, and knowledge sharing.
- Troubleshoot production data and reporting issues by tracing source feeds, transformation logic, SQL behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Partnered with infrastructure, application, analytics, and business teams to translate capacity and performance signals into practical recommendations for operational stability and service delivery.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring analytics for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built operational dashboards and reporting views that converted raw monitoring data into actionable reliability and performance insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise CA APM telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents across large financial-services infrastructure.
- Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry, replacing manual exports with repeatable reporting workflows.
- Established dashboard, alert, SLA threshold, reporting, and technical documentation standards adopted across client environments.
- Mentored technical teams and partnered with J2EE, WebLogic, infrastructure, database, and operations groups to analyze telemetry, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and data feed processes for telecom-scale transaction systems.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.
- Implemented business logic in billing and integration workflows using C/C++, Pro*C, PL/SQL, SQL scripts, XML, shell automation, and enterprise messaging interfaces.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

HorizonScale — AI Capacity Forecasting Engine

Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.

Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Capacity Forecasting

Enterprise APM Reporting Automation

Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.

Technologies: KornShell/ksh, Perl, Unix/Linux, CA APM, SQL, Automation, Monitoring

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw infrastructure telemetry ingestion, transformation, curated analytics datasets, query optimization, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL